End-to-End Machine Learning Assignment

Project Title:

Diamond Price Prediction using KNN and Streamlit Deployment

Objective:

Develop a complete end-to-end machine learning application to predict diamond prices using the K-Nearest Neighbors (KNN) regression algorithm. The project should include model development, evaluation, and deployment using Streamlit Cloud. Finally, submit the deployed application link.

Dataset:

- `diamonds.csv`
- `description.txt`

Part 1: Machine Learning Pipeline

Perform the following steps:

1. Load the dataset using Pandas.
2. Define the input features (X) and the target variable (`price`).
3. Split the dataset into training and testing sets using a 75:25 ratio.
4. Apply data preprocessing:
    - Encode categorical features
    - Scale numerical features
5. Train a KNN Regressor model using Scikit-learn.
6. Evaluate the model using the following metrics:
    - Mean Absolute Error (MAE)
    - Root Mean Squared Error (RMSE)
    - R^2 Score
7. Save the trained pipeline using Pickle for future use.